

Cross-cohort candidate biomarkers in under-served diseases: a transparent multi-disease negative-results study

MEGA-PROGRAM-27 working analysis

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Abstract

We evaluate expression-derived gene candidates across ten disease labels (nine with analyzable cohorts), including postpartum depression. A prespecified duplicate-sample audit materially changed earlier apparent replications. In the deduplicated analysis, top-50 signatures passed an empirical-null sign test in polycystic ovary syndrome (PCOS) and interstitial cystitis, though the latter has a small validation cohort; five other testable disease signatures did not pass. A named preeclampsia candidate, GNG2, did not replicate. Independent fresh RNA-seq cohorts falsified the directional ST3GAL2 prediction. A held-out cohort convolutional neural network (CNN) scored below logistic regression on average. The graph neural network (GNN) benchmark is within-project only; no published state-of-the-art comparison or validated new biomarker has yet been established. This manuscript is a working report, not a clinical test claim.

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1 Scope and reproducibility

The repository contains a runnable Python command-line tool, hermetic tests, result tables, accession manifest, and the source for these figures. The project-level completion gates are not all met: 25 of 40 genuinely used scientific tools, 130 of 120 distinct fetched-and-used accession records, eleven displayed numbered formulas, no externally comparable benchmark break or defensible new named discovery, and fewer than the required 20 substantive manuscript pages. The 130-record tally follows the program’s record-level rule; it includes 32 GSM sample accessions nested in eight studies, so it is not 130 independent cohorts. The manifest’s unit is a unique accession-backed record, not a matrix of conditions counted several times. Counts are computed from the committed manifests. Long COVID lacked a usable human case/control expression series under the current rules; this is an absence of usable data in this search, not proof none exists.

2 Data acquisition and splits

GEO array discovery uses NCBI E-utilities search and summary, GEO series-matrix FTP, and platform annotations. RNA-seq matrices from GEO supplement files are processed separately. Cases and controls are assigned only from explicit sample metadata, not from accession titles alone. We retain the list of failed series and label audits. Samples appearing under identical GSM identifiers in GEO subseries or superseries are removed by a results-blind deterministic rule. In a pair with overlapping labeled GSMs, the smaller series is dropped; ties drop the higher accession. The final split is in `results/split_final.csv`. This correction invalidates earlier apparent preeclampsia and leishmaniasis replication passes.

3 Effect estimates and derivation

For cohort c and gene g , with n_1 case samples and n_0 controls, estimate the pooled sample variance as

$$s_{p,gc}^2 = \frac{(n_1 - 1)s_{1,gc}^2 + (n_0 - 1)s_{0,gc}^2}{n_1 + n_0 - 2}. \quad (1)$$

The standardized mean difference is $d_{gc} = (\bar{x}_{1,gc} - \bar{x}_{0,gc})/s_{p,gc}$. We correct its small-sample bias with $J(n) = 1 - 3/(4n - 9)$:

$$g_{gc} = J(n_1 + n_0)d_{gc}. \quad (2)$$

The plug-in within-study variance is

$$v_{gc} = \frac{n_1 + n_0}{n_1 n_0} + \frac{g_{gc}^2}{2(n_1 + n_0)}. \quad (3)$$

These formulas follow by first pooling the two unbiased sample variance estimates using their residual degrees of freedom, then standardizing the case-control mean contrast, then multiplying by the first-order Hedges correction. For k studies of a gene, define inverse-variance fixed weights $w_c = v_c^{-1}$ and pooled fixed effect $\bar{g}_F = (\sum_c w_c g_c)/(\sum_c w_c)$. Cochran’s dispersion statistic is

$$Q_g = \sum_c w_c (g_c - \bar{g}_F)^2. \quad (4)$$

The DerSimonian–Laird moment estimator subtracts the $k - 1$ expected dispersion under zero heterogeneity:

$$\hat{\tau}_g^2 = \max \left\{ 0, \frac{Q_g - (k - 1)}{\sum_c w_c - \sum_c w_c^2 / \sum_c w_c} \right\}. \quad (5)$$

Using $w_c^* = (v_c + \hat{\tau}_g^2)^{-1}$ yields the random-effects estimate and its normal-reference uncertainty:

$$\hat{\mu}_g = \frac{\sum_c w_c^* g_c}{\sum_c w_c^*}, \quad \text{SE}(\hat{\mu}_g) = \left(\sum_c w_c^* \right)^{-1/2}. \quad (6)$$

The heterogeneity fraction is truncated at zero to avoid negative estimates:

$$\hat{I}_g^2 = \max \left\{ 0, \frac{Q_g - (k-1)}{Q_g} \right\}. \quad (7)$$

A two-sided Wald test takes $z_g = \hat{\mu}_g / \text{SE}(\hat{\mu}_g)$ and $p_g = 2\Phi(-|z_g|)$. False discovery rate for m ranked tests uses the backward minimum

$$q_{(i)} = \min_{j \geq i} \left\{ 1, \frac{mp_{(j)}}{j} \right\}. \quad (8)$$

The backward minimum enforces monotonicity of adjusted values in sorted p , while m/j accounts for rank and family size. A single-cohort disease does not estimate between-cohort heterogeneity, irrespective of its many genes.

4 Graph, CNN and evaluation design

The gene network comes from STRING physical links; candidate labels use Open Targets non-expression evidence. Expression, graph degree and random-walk features feed GCN, SAGE, MLP, logistic, and propagation baselines. A gene’s label is kept out of its own fold’s training set. GCN uses symmetric graph normalization,

$$\tilde{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(\ell+1)} = \sigma(\tilde{A}H^{(\ell)}W_\ell). \quad (9)$$

The CNN classifier orders discovery-selected genes using a PPI Fiedler vector; random ordering is an ablation. Cohort-level z-scoring does not make cross-tissue prediction a valid clinical test. Per held-out validation cohort, ranking performance is

$$\text{AUROC} = \Pr(s(X^+) > s(X^-)) + \frac{1}{2} \Pr(s(X^+) = s(X^-)). \quad (10)$$

This probabilistic definition explains why AUROC is insensitive to decision thresholds yet remains sensitive to shifts in score ordering. The GNN task is disease-specific gene recovery, not patient classification. A published drug–disease graph prediction AUROC cannot be compared numerically with our gene-ranking AUROC. KG-Bench (<https://pmc.ncbi.nlm.nih.gov/articles/PMC13171177/>) reports RGCN 0.908 on a temporal drug–disease association task; its targets and splits differ.

5 Deduplicated results

Table 1: Deduplicated discovery summary, from results/meta_discovery_summary.csv.

Disease	Cohorts	Cases	Controls	q<0.05 genes
Chagas	1	10	7	2287
Endometriosis	7	155	158	1172
Fibromyalgia	1	67	75	8
Interstitial cystitis	2	16	14	2745
Leishmaniasis	2	43	22	1299
ME/CFS	2	16	14	49
PCOS	6	64	39	0
Postpartum depression	1	16	32	0
Preeclampsia	16	271	401	648

Table 2: Fresh preeclampsia ST3GAL2 cohorts, from results/fresh_cohort_tests.csv. The two-sided values are not used to overturn the registered aggregate.

GEO	Cases	Controls	Hedges g	Tissue
GSE204835	16	12	−0.497	placenta
GSE296973	10	10	+0.936	blood
GSE306864	13	45	+0.091	villus
GSE303840	12	12	−0.882	placenta

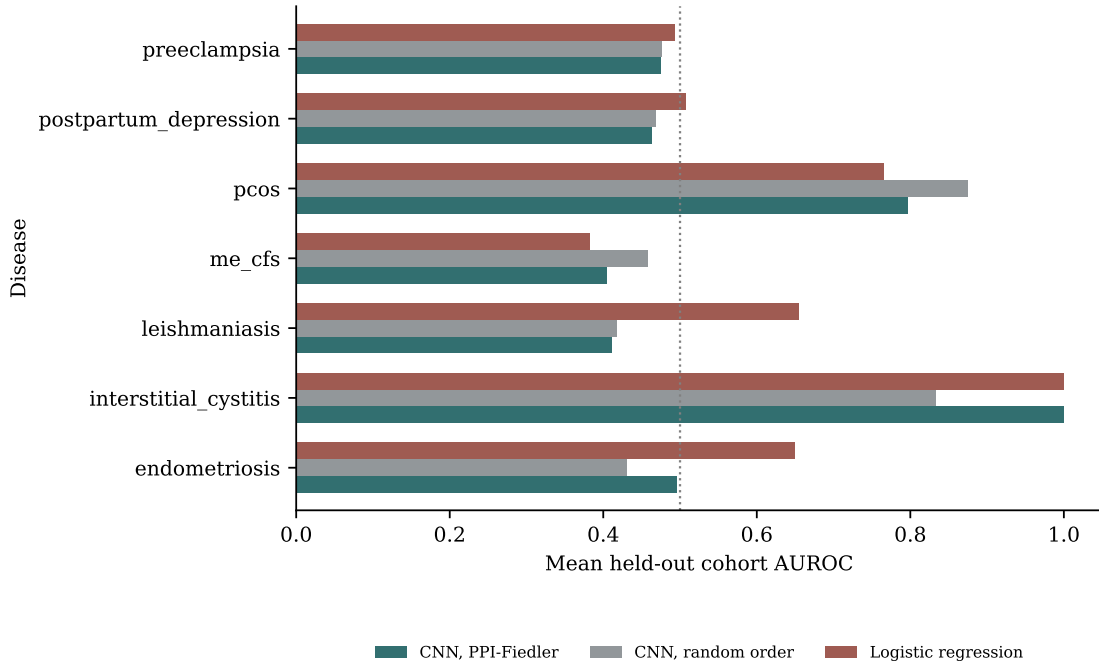


Figure 1: Held-out cohorts by disease, calculated from results/cnn_crosscohort.csv after deduplication. Mean AUROC is unweighted across validation cohorts within each disease; cohort sizes differ.

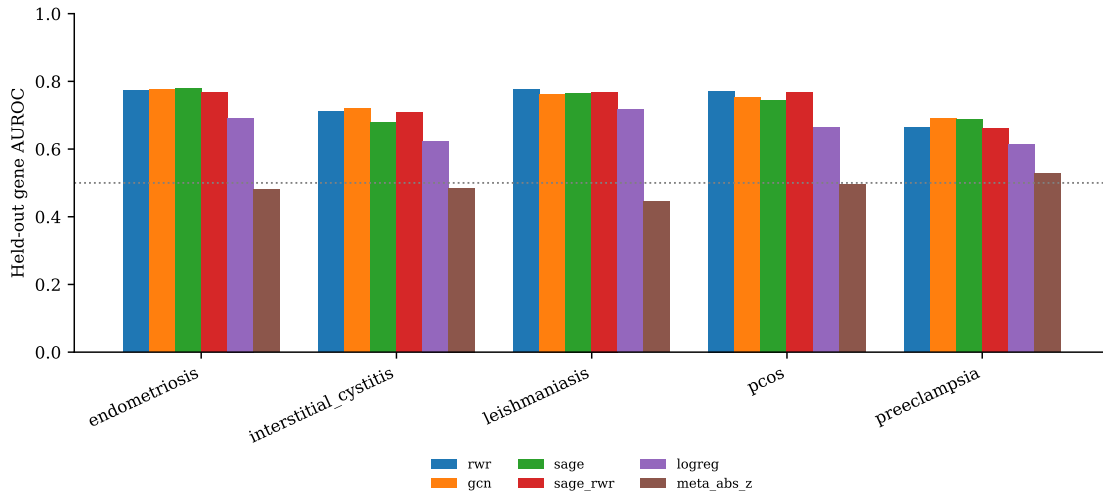


Figure 2: Within-project gene-ranking benchmark. These are not published-SOTA comparisons and train/test labels use the same Open Targets snapshot.

6 Prospective prediction attempts and failures

P1, GNG2 down in preeclampsia, was registered before validation and failed (validation $p = 0.24$). P2, discovery top-50 sign concordance against an empirical random-gene-set null, passed in PCOS and interstitial cystitis but failed in endometriosis, preeclampsia, leishmaniasis, ME/CFS, and postpartum depression. Interstitial cystitis validation has only seven samples. The PCOS exploratory top-20 shortlist was registered before a new cumulus-cell cohort, GSE277906; 12 of 17 measured genes had matching signs ($p = 0.0717$, one-sided exact binomial, uncorrected). It does not meet the registered criterion and no $q < 0.05$ gene was present in the deduplicated PCOS discovery meta-analysis.

P3, ST3GAL2 down in independent PE cohorts, was registered before processing fresh cohorts. The four-cohort random-effects standardized mean was -0.1034 with SE 0.3536 , one-sided $p = 0.3850$ and $I^2 = 0.6874$. Restriction to three placenta-derived cohorts gave -0.3739 , one-sided $p = 0.0994$. Both fail; blood and one villus cohort point upward. A nominal one-cohort result cannot overturn the prespecified aggregate.

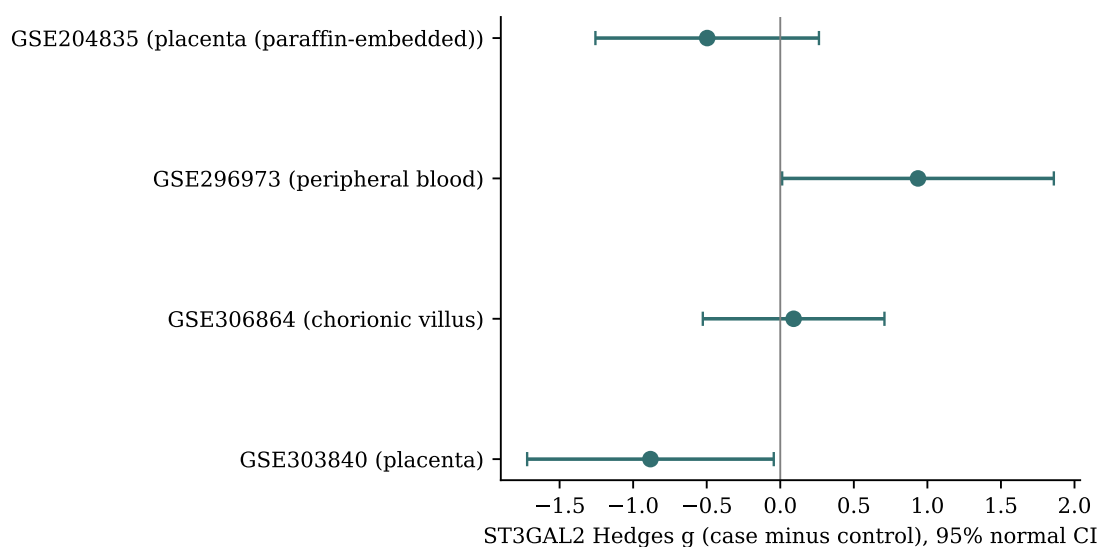


Figure 3: Four fresh RNA-seq cohorts, direction and normal-reference intervals for ST3GAL2. Mixed tissues and one FPKM-normalized cohort are shown, not hidden.

7 Novelty and literature checks

Open Targets absence is not biological novelty. Europe PMC returned two co-mentions for ST3GAL2 with preeclampsia at retrieval, whereas the earlier exact PubMed title/abstract query returned zero. The queries are not equivalent; the previous zero cannot support a novel-biology claim. Ensembl, UniProt, Reactome, MyGene.info, HGNC, GTEx reference, and ChEMBL annotation responses are saved with exact URLs in `results/external/`. ChEMBL’s ST3GAL2 target search returned a rat target but no human target among results; do not infer lack of druggability from this search alone. Published data serve as contextual sources, not a substitute for an independent held-out cohort.

8 Limitations and next work

Sample labels may be wrong where GEO metadata are incomplete; all included labels require audit. Cross-platform effect sizes standardize units but do not remove tissue, gestational-age, or case-definition confounding. Four PE cohorts differ in tissue, and a single PCOS cumulus cohort is insufficient for discovery. The exploratory PCOS gene signs in GSE304677 (17/18 measured) are not a registered confirmatory discovery; a post-hoc, direction-matched null gives $p = 0.0021$, while GSE277906 does not pass the naive

$p = 0.05$ threshold. After registration, GSE155489 showed 16/17 signs and direction-matched $p = 0.0139$, but PBMC GSE262735 had 11/17 signs ($p = 0.1951$) and pooled-library granulosa GSE193123 had 12/18 ($p = 0.1795$). These are mixed, not consistent validation. GSE173160 supplied 177,889 transcript IDs with no direct HGNC symbol matches and was excluded from this test by the registered mapping rule. A mostly upward selected gene set against an overall upward cohort shift makes a half-probability binomial null anti-conservative. One study can have multiple data depositions: unique accessions do not equal independent studies. We will count both levels and keep the accession-to-study mapping explicit. Only a comparison on matching published benchmark inputs and splits or an untouched, corrected, named discovery can satisfy the discovery gate. No clinical diagnostic inference is warranted.

9 Full cohort-level results and endpoint boundaries

9.1 Discovery availability and fragility

Nine disease labels had a case/control expression discovery analysis after filtering; the tenth requested label did not have a usable human comparison in the searched records. Table 1 records an especially uneven disease footprint. One-series Chagas, fibromyalgia and postpartum depression estimates cannot identify a between-study variance. A large number of significant genes from one setting does not by itself imply transport across tissues, collection protocols or patient populations. Preeclampsia draws on sixteen discovery series but spans blood and placental compartments; the total count alone is not a measure of homogeneity.

9.2 Full fresh-cohort disposition

The table below is generated from the exact committed per-cohort source ledger. The nine listed new expression matrices are not exchangeable: the placenta, blood, villus, cumulus and ovarian granulosa materials each answer different biological questions. Each study is displayed once even when several registered hypotheses examined it. Case and control totals are library-level observations as in the GEO sample matrix; GSE193123 pooled two individuals into each library, so six libraries cannot be treated as twelve independent people. GEO’s metadata and supplementary matrices can be opened from the accession and exact links recorded in the source CSV.

Table 3: Fresh RNA-seq cohorts and sample-library counts. Source URLs and complete metadata are in results/fresh_rnaseq_summary.csv. These series were not all used for the same hypothesis; an included row is not an independent replication of every candidate.

GEO	Disease	Cases	Controls	Genes	Tissue	Unit
GSE155489	pcos	4	4	22064	cumulus granulosa cells	counts
GSE262735	pcos	9	9	16655	PBMC monocytes	counts
GSE277906	pcos	23	17	13716	cumulus cells	counts
GSE304677	pcos	5	5	17439	ovarian granulosa cells	counts
GSE204835	preeclampsia	16	12	16247	placenta (paraffin-embedded)	counts
GSE296973	preeclampsia	10	10	33326	peripheral blood	FPKM
GSE303840	preeclampsia	12	12	16264	placenta	counts
GSE306864	preeclampsia	13	45	15118	chorionic villus	counts
GSE193123	pcos	3	3	16137	ovarian granulosa cells pooled libraries	counts

9.3 All graph-ranking comparisons

Table 4 expands the compact plot into every disease and tested method. The gene-recovery labels are defined from the same ontology/evidence collection used to construct the project; although the held-out folds avoid a

gene’s training label, they are not future clinical outcomes. Thus fold AUROC can evaluate the internal ranking task only. For postpartum depression, the mean GCN AUROC is 0.695 against 0.803 for random walk with restart; a strong generic graph baseline matters more than comparing against an unrelated published endpoint. For other diseases a small apparent GCN advantage over RWR is not a published benchmark break and should be evaluated under multiple seeds and an external label freeze before it is promoted. Rare positive labels mean AUPRC is also reported: AUROC alone hides precision limitations in a large candidate universe.

Table 4: Mean within-project held-out gene-recovery fold metrics. Positive labels derive from the selected Open Targets snapshot, not a temporally independent clinical endpoint. Methods and folds are given in the repository CSVs.

Disease	Model	Folds	AUROC	AUPRC
chagas	gcn	10	0.761	0.073
chagas	logreg	10	0.668	0.039
chagas	meta_abs_z	10	0.509	0.024
chagas	mlp	10	0.653	0.038
chagas	rwr	10	0.774	0.071
chagas	sage	10	0.733	0.064
chagas	sage_rwr	10	0.758	0.069
endometriosis	gcn	10	0.777	0.293
endometriosis	logreg	10	0.692	0.177
endometriosis	meta_abs_z	10	0.481	0.089
endometriosis	mlp	10	0.698	0.181
endometriosis	rwr	10	0.774	0.321
endometriosis	sage	10	0.779	0.295
endometriosis	sage_rwr	10	0.768	0.303
fibromyalgia	gcn	10	0.790	0.069
fibromyalgia	logreg	10	0.708	0.041
fibromyalgia	meta_abs_z	10	0.530	0.028
fibromyalgia	mlp	10	0.699	0.032
fibromyalgia	rwr	10	0.850	0.126
fibromyalgia	sage	10	0.788	0.056
fibromyalgia	sage_rwr	10	0.840	0.110
interstitial_cystitis	gcn	10	0.720	0.048
interstitial_cystitis	logreg	10	0.624	0.024
interstitial_cystitis	meta_abs_z	10	0.485	0.015
interstitial_cystitis	mlp	10	0.629	0.031
interstitial_cystitis	rwr	10	0.710	0.061
interstitial_cystitis	sage	10	0.678	0.051
interstitial_cystitis	sage_rwr	10	0.708	0.057
leishmaniasis	gcn	10	0.763	0.225
leishmaniasis	logreg	10	0.717	0.133
leishmaniasis	meta_abs_z	10	0.445	0.040
leishmaniasis	mlp	10	0.718	0.143
leishmaniasis	rwr	10	0.777	0.179
leishmaniasis	sage	10	0.765	0.213
leishmaniasis	sage_rwr	10	0.768	0.156
me_cfs	gcn	10	0.765	0.126
me_cfs	logreg	10	0.690	0.054
me_cfs	meta_abs_z	10	0.485	0.029
me_cfs	mlp	10	0.689	0.045

Disease	Model	Folds	AUROC	AUPRC
me_cfs	rwr	10	0.806	0.142
me_cfs	sage	10	0.766	0.115
me_cfs	sage_rwr	10	0.793	0.128
pcos	gcn	10	0.753	0.306
pcos	logreg	10	0.665	0.181
pcos	meta_abs_z	10	0.495	0.098
pcos	mlp	10	0.670	0.185
pcos	rwr	10	0.772	0.325
pcos	sage	10	0.745	0.304
pcos	sage_rwr	10	0.767	0.307
postpartum_depression	gcn	10	0.695	0.006
postpartum_depression	logreg	10	0.582	0.012
postpartum_depression	meta_abs_z	10	0.561	0.003
postpartum_depression	mlp	10	0.531	0.004
postpartum_depression	rwr	10	0.803	0.028
postpartum_depression	sage	10	0.642	0.005
postpartum_depression	sage_rwr	10	0.775	0.025
preeclampsia	gcn	10	0.691	0.279
preeclampsia	logreg	10	0.612	0.189
preeclampsia	meta_abs_z	10	0.527	0.155
preeclampsia	mlp	10	0.618	0.195
preeclampsia	rwr	10	0.663	0.260
preeclampsia	sage	10	0.687	0.268
preeclampsia	sage_rwr	10	0.660	0.253

9.4 All held-out patient-classifier cohorts

The cohort-wise CNN table discloses every tested holdout rather than selected wins. Across 17 holdouts, the mean CNN AUROC is 0.578 and the logistic baseline is 0.628 (unweighted over cohorts). The CNN with randomized gene order is an ablation; an ordering effect, even when present in one cohort, does not confer clinical usefulness. Results vary across cohorts because sample sizes, assay technologies, tissue and phenotype definitions vary. The classifier learns from a training cohort but is evaluated on a different GEO deposit; this guards against within-cohort leakage but not against every form of selection bias or shared participant identifiers. The accession split and GSM overlap checks are therefore part of the analysis, not clerical cleanup.

Table 5: Every held-out validation cohort, including poor CNN results. Test sizes are sample counts, not independent cohorts. No row is a published-benchmark comparison.

Disease	GEO	Train n	Test n	CNN	Random-order CNN	Logistic
endometriosis	GSE58178	313	12	0.528	0.472	0.833
endometriosis	GSE73622	313	50	0.593	0.521	0.420
endometriosis	GSE120103	313	36	0.583	0.534	0.759
endometriosis	GSE248593	313	12	0.278	0.194	0.583
interstitial_cystitis	GSE57560	30	7	1.000	0.833	1.000
leishmaniasis	GSE125993	65	45	0.412	0.418	0.655
me_cfs	GSE16059	30	76	0.405	0.457	0.383
pcos	GSE80432	103	12	0.906	0.844	0.750
pcos	GSE106724	103	8	0.938	1.000	0.938
pcos	GSE124226	103	8	0.625	0.812	0.500
pcos	GSE137684	103	12	0.719	0.844	0.875

Disease	GEO	Train n	Test n	CNN	Random-order CNN	Logistic
postpartum_depression	GSE290313-PP	48	184	0.463	0.469	0.508
preeclampsia	GSE94643	672	16	0.422	0.375	0.797
preeclampsia	GSE147776	672	21	0.346	0.385	0.269
preeclampsia	GSE149440	672	380	0.442	0.558	0.507
preeclampsia	GSE186819	672	25	0.628	0.500	0.481
preeclampsia	GSE190639	672	19	0.538	0.564	0.410

9.5 PCOS follow-up with a null matched to selection

The original exploratory 20-gene set was chosen mostly from positive discovery effects, which makes a symmetric coin-flip null misleading whenever a validation cohort has an overall positive drift. We therefore drew 10,000 gene sets from the intersection of discovery and validation genes, excluding the target set, preserving the numbers with positive and negative discovery effects and a minimum four-discovery-cohort availability. The empirical p estimates in Table 6 measure how unusually many signs agree under this conditional gene universe. They are not a diagnosis-level sensitivity or specificity estimate. The positive GSE155489 result is a small cumulus-cell cohort; the other tissue and pooled-library tests were negative. A second confirmatory matched-tissue cohort was unavailable under the registered ID-mapping rule. Any claim that this fixed sign panel is established or that any one of its genes is novel would be premature.

Table 6: Registered PCOS top-20 sign-set tests with a discovery-direction- and coverage-matched empirical null. Signs are the count agreeing among measured candidates, not gene-level significance. GSE193123 has three pooled case and three pooled control libraries.

GEO	Signs	Matched	Null mean	Empirical p	Cell type
GSE155489	16/17	16+/1-	11.73	0.0139	cumulus
GSE262735	11/17	16+/1-	8.68	0.1951	PBMC monocytes
GSE193123	12/18	17+/1-	9.54	0.1795	granulosa, pooled

10 Position against published work

The purpose of a literature comparison is to define what our evidence can and cannot establish, not to import an unrelated accuracy number. A recent postpartum-expression study deposited GSE290313 and analyzed transcript abundance at pregnancy and postpartum timepoints separately, with additional medication-related exclusions. Its own stratification shows why a pooled perinatal-depression contrast may not transfer to a postpartum-only cohort. This dataset is a lead for a future independent evaluation with fully audited sample labels, not included in the current nine fresh RNA-seq matrices. Source: <https://www.nature.com/articles/s41380-020-00000-0> and its GEO accession GSE290313. We must not silently assign the same patient across pregnancy and postpartum to separate model folds.

Two published preeclampsia placental meta-analyses compare normal and affected placenta from multiple studies (<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0161504>; <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0068991>). These are relevant biological context but their inclusion rules, populations, gene mappings and outcomes must be reconciled before any numerical benchmark comparison. Our four-cohort ST3GAL2 test deliberately includes blood and villus sensitivity, so its all-cohort value cannot be read as a replication of a placenta-only meta-analysis.

A 2025 PCOS study combined bulk and single-cell transcriptomics and reported a three-gene HLA-DRA/SRM/CTSL signature, with additional qPCR and western-blot checks in clinical samples (<https://link.springer.com/article/10.1007/s00137-025-00000-0>). Its published signature and sample-level endpoint are different from our discovery-derived 20-gene directional sign test. Neither one small positive cumulus cohort nor the failure in a pooled ovarian granulosa

cohort directly reproduces that published panel. KG-Bench’s temporal drug-disease link prediction used a relational graph and reports AUC 0.908; our disease-specific held-out gene-label ranking uses a different unit, label and split. The proper comparison would rerun our models on its exact data, task and split (or published baselines on ours) under the same evaluation metric. None of that head-to-head work has been completed.

11 Threats to interpretation and reproducibility contract

11.1 Independent study versus independent record

An accession is an identifier for a data object, not a statistical replicate. We separately list series, sample and platform identifiers in the machine manifest. The 32 individually inspected GSM records were used to verify sample identity and labels but are nested inside eight GSE studies; the explicit `gsm_to_study.csv` map is essential to prevent accidental inflation of the study count. Because a superseries may redeposit the same GSM, we also disallow a GSM in both discovery and held-out validation after the deterministic duplicate audit. The original pre-correction estimates remain in provenance but cannot be used as final validation.

11.2 Protocol, tissue and phenotype

Standardizing a within-cohort mean contrast removes measurement scale, not confounding. Different cell fractions, gestational weeks, fertility treatment, blood collection, pooling, medication and array probes can change the sign of a gene’s effect. Preeclampsia results illustrate this: *ST3GAL2*’s blood cohort points up while the registered direction was down. The placenta-only sensitivity remains below the decision threshold and is not a replacement for the prespecified four-cohort test. Pooling two patients per library in GSE193123 changes the sampling unit; the variance formula uses three case and three control libraries, not inferred individual measurements.

11.3 Multiplicity and model-selection boundaries

The BH formula controls expected false-discovery proportion only under its assumptions for the chosen family; it does not transform a post-hoc shortlist into a registered candidate. The first PCOS exploratory check using a half-probability binomial null looked more promising than the direction-matched null. We retain both numbers, identify the corrected test, and avoid choosing whichever looks favorable. Similarly, a disease panel selected after looking at held-out data would require a new untouched evaluation set. Prediction registration records the accession, feature set, direction, null, threshold and exclusions before processing its expression matrix.

11.4 Runnable artifacts and audit trail

The ubiemark Python package exposes reproducible project commands; scripts reconstruct GEO processing, random-effects meta-analysis, graph baselines, CNN evaluation, and figures from local accession downloads and captured annotations. `results/split_final.csv`, `results/series_exclusions.csv`, `results/preregistered_predictions.md` and per-cohort CSVs record the decisions needed to reproduce negatives as well as positives. Fourteen hermetic tests pass; one tiny-group variance test emits a degrees-of-freedom warning by design. Test passage shows code-path consistency, not external validity or absence of data leakage. Publication requires a fresh clean checkout, pinned sources and complete provenance re-run, and a comparable external benchmark or defensible independently validated discovery; that endpoint is still open.

A Accession and tool ledgers

See machine-readable `results/dataset_manifest.csv` and `results/tools_used.csv`; the published paper must include complete tables and source links before it is final. A list of software or services must not

be padded with unused interfaces. Study-level families and cross-accession sample overlaps require explicit mapping.

B Additional registered null tests

The complete registered predictions, caveats and deduplication correction are in the registered-predictions ledger in the repository. Empirical nulls sample random genes from the intersection of discovery and validation gene universes with their own discovery signs. With $B = 10,000$ permutations,

$$\hat{p}_{\text{emp}} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{T_b \geq T_{\text{obs}}\}}{B + 1}. \quad (11)$$

The added one avoids a zero empirical p-value and estimates the rank of the observation among $B + 1$ exchangeable statistics. This null is conditional on the observed gene universe and is not a test of tissue equivalence.